

Chapter 13: Motion and Time

Subject	Science (NCERT Class 7)
Chapter	13 – Motion and Time
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors are close-but-wrong: average speed taken as a plain mean of two speeds, unit conversions multiplied the wrong way, pendulum period blamed on bob mass or amplitude, distance-time graph slope confused with the y-value, speedometer/odometer swapped, and rest segments counted as motion. Do not write on this paper — mark answers on your answer sheet.

1. A point on the second-hand tip of a wall clock keeps tracing the same circle, returning to the 12 mark every 60 s. The two descriptors that fit this motion best are:
(A) rectilinear and oscillatory
(B) oscillatory and non-periodic
(C) circular and periodic
(D) random and rotational
2. A go-kart covers 240 m in 16 s at a steady rate. Its speed is:
(A) 16 m/s
(B) 15 m/s
(C) 0.067 m/s
(D) 3840 m/s
3. Convert a speed of 108 km/h into m/s.
(A) 388.8 m/s
(B) 30 m/s
(C) 60 m/s
(D) 10.8 m/s
4. A cyclist rides 12 km in 30 minutes, then rests for 30 minutes, then rides another 12 km in 1 hour. The average speed for the whole journey is:
(A) 18 km/h
(B) 16 km/h
(C) 24 km/h
(D) 12 km/h

- 5.** A car covers the first half of a trip at 30 km/h and the second half (equal distance) at 60 km/h. Its average speed for the whole trip is:
- (A) 45 km/h (B) 50 km/h
(C) 90 km/h (D) 40 km/h
- 6.** Two simple pendulums hang side by side. One has a heavy iron bob, the other a light wooden bob, but both strings are exactly the same length and both are released from the same small angle. Their time periods are:
- (A) longer for the heavy iron bob
(B) equal, because period depends only on length
(C) longer for the light wooden bob (D) longer for whichever is pulled back farther
- 7.** A driver glances at her dashboard and reads 'fast 80', then notices another number that has climbed by 45 since she set out. The instruments she is reading are, respectively:
- (A) the odometer (current speed) and the speedometer (distance travelled)
(B) two speedometers showing speed twice
(C) the speedometer (current speed) and the odometer (distance travelled)
(D) a clock and a speedometer
- 8.** Convert a speed of 25 m/s into km/h.
- (A) 6.94 km/h (B) 75 km/h
(C) 90 km/h (D) 45 km/h
- 9.** An athlete runs 400 m in 50 s, slows to a walk and covers the next 100 m in 100 s. His average speed over the full 500 m is:
- (A) about 3.3 m/s (B) 4.5 m/s
(C) 5 m/s (D) 1 m/s
- 10.** On a distance-time graph an object's line is a straight slant from (0 s, 0 m) up to (5 s, 40 m), then a horizontal line from (5 s, 40 m) to (9 s, 40 m). During the second part the object is:
- (A) moving faster, because the line is now flat
(B) moving at 40 m/s, read straight off the graph
(C) moving backward toward the start
(D) at rest, since the distance does not change with time
- 11.** A car runs at 80 km/h for 1 hour and then at 40 km/h for 3 hours. Its average speed for the whole trip is:
- (A) 60 km/h (B) 55 km/h
(C) 50 km/h (D) 120 km/h
- 12.** A drilling machine's bit spins about a fixed central axis without moving from its spot. This kind of motion is best called:

- (A) rectilinear motion (B) rotational motion
(C) oscillatory motion (D) translational motion in a straight line

13. On a distance-time graph, line P passes through (0,0) and (4 s, 24 m); line Q passes through (0,0) and (4 s, 16 m). Which object is faster, and by how much?

- (A) Q is faster; the lower line is always quicker
(B) they are equally fast; both reach 4 s together
(C) P is faster by 8 m/s (D) P is faster; P is 6 m/s, Q is 4 m/s

14. A bicycle wheel valve clicks past a fixed sensor once per turn, and the clicks come exactly 0.5 s apart at steady speed. Using these regular clicks to mark off equal time intervals works because the motion is:

- (A) non-uniform, which is why we can time it
(B) periodic, so equal time intervals repeat reliably
(C) random, giving an average over many clicks (D) oscillatory but not periodic

15. A motorcycle travels at a steady 90 km/h. How many metres does it cover in 8 seconds?

- (A) 720 m (B) 200 m
(C) 25 m (D) 180 m

16. A pendulum is set swinging and released from a large angle one day and a small angle the next, with the string length kept identical. Compared with the small-angle day, the time period on the large-angle day is:

- (A) the same, because amplitude does not change the period
(B) longer, because it swings through a bigger arc
(C) shorter, because it moves faster (D) longer, because it carries more energy

17. On a distance-time graph an object's line passes through (1 s, 5 m) and (6 s, 35 m). Its speed is:

- (A) 5 m/s (B) 7 m/s
(C) 30 m/s (D) 6 m/s

18. A bus covers a journey in three equal time-segments of 1 hour each, at 36 km/h, 54 km/h and 72 km/h. Its average speed for the whole trip is:

- (A) 162 km/h (B) 48 km/h
(C) 54 km/h (D) 60 km/h

19. A leopard sprints at a steady 20 m/s. How long does it take to cover 360 m, and what is that speed in km/h?

- (A) 18 s and 72 km/h (B) 18 s and 56 km/h
(C) 20 s and 72 km/h (D) 7200 s and 72 km/h

20. An ant covers 9 cm in 3 s. Expressed in metres per second, its speed is:

(A) 3 m/s

(B) 0.3 m/s

(C) 27 m/s

(D) 0.03 m/s

21. On a distance-time graph, the line for car X is steeper than the line for car Y, and both are straight slants from the origin. This tells you that:

(A) X has travelled less total distance than Y (B) X is moving faster than Y

(C) Y is moving faster than X

(D) X started later than Y

22. A clock's pendulum completes one full back-and-forth swing (one oscillation) in 2 s. How many complete oscillations does it make in 5 minutes?

(A) 150

(B) 300

(C) 75

(D) 2.5

23. On a distance-time graph, two straight lines from different starting heights cross at the point (4 s, 20 m). At that instant the two objects:

(A) are at the same distance from the start (B) have the same speed

(C) both come to rest

(D) reverse their directions

24. A pendulum makes 24 complete oscillations in 36 s. Its string is then shortened so that it makes 24 oscillations in only 18 s. What happened to its time period?

(A) it doubled, from 1.5 s to 3 s

(B) it stayed the same at 1.5 s

(C) it halved, from 1.5 s to 0.75 s

(D) it became 24 s, the number of swings

25. Convert a speed of 18 km/h into m/s.

(A) 5 m/s

(B) 64.8 m/s

(C) 0.5 m/s

(D) 50 m/s

26. A motorist plans a 300 km trip at an average of 75 km/h. After the first 150 km, covered at 50 km/h, what average speed must she keep over the remaining 150 km to still finish at the planned 75 km/h overall?

(A) 100 km/h

(B) 75 km/h

(C) 150 km/h

(D) 125 km/h

27. Two cars cover the same distance. Car A takes 4 hours and car B takes 5 hours. Which is faster, and in what ratio are their speeds (A : B)?

(A) B is faster; speeds are in the ratio 5 : 4 (B) A is faster; speeds are in the ratio 5 : 4

(C) A is faster; speeds are in the ratio 4 : 5

(D) they are equally fast since the distance is the same

28. An object's distance-time graph is a straight slanted line passing through (0,0) and (3 s, 12 m). Its motion is best described as:

(A) non-uniform motion at 4 m/s

(B) uniform motion at 12 m/s

(C) uniform motion at 4 m/s

(D) at rest the whole time

29. On a distance-time graph an object's line slopes up from (0 s, 0 m) to (4 s, 40 m), stays horizontal from (4 s, 40 m) to (7 s, 40 m), then slopes up again from (7 s, 40 m) to (10 s, 70 m). The total distance travelled in the 10 s is:

- (A) 70 m
- (B) 150 m
- (C) 110 m
- (D) 40 m

30. On a distance-time graph, an object's line gets steeper and steeper as time goes on (it curves upward). The object is:

- (A) slowing down, because the curve is bending
- (B) moving at a constant speed
- (C) speeding up (moving faster as time passes)
- (D) at rest, because it is one continuous line

31. A sprinter runs 150 m in 15 s. A galloping horse covers 600 m in 1 minute. Comparing their speeds:

- (A) they are equally fast, both at 10 m/s
- (B) the sprinter is faster
- (C) the horse is faster
- (D) the horse is faster by 590 m/s

32. To find a car's true average speed over a complete trip, the most direct instruments to use are:

- (A) the odometer (for total distance) and a clock (for total time)
- (B) the speedometer alone, read once at the end
- (C) the speedometer alone, read once at the start
- (D) a clock alone

33. A wind-up toy car drives 6 m in the 1st second, 4 m in the 2nd second, and 2 m in the 3rd second along a straight line. Its motion is:

- (A) uniform, because it keeps moving forward
- (B) non-uniform, slowing down
- (C) non-uniform, speeding up
- (D) at rest, because it is on the floor

34. Light travels about 300,000 km in 1 s. About how far does it travel in 4 s?

- (A) 75,000 km
- (B) 300,000 km
- (C) 1,200,000 km
- (D) 1,200 km

35. Two pendulums hang together. Pendulum A is 1 m long and pendulum B is much longer at 2 m. Over one minute, which one completes more oscillations?

- (A) B, because a longer string swings more often
- (B) A, because the shorter pendulum has the smaller time period
- (C) they complete the same number, since mass is equal
- (D) B, because it carries more energy

36. On a distance-time graph, line A goes from (0,0) to (10 s, 100 m) and line B goes from (0,0) to (10 s, 60 m). The ratio of speed of A to speed of B is:

- (A) 5 : 3
- (B) 3 : 5
- (C) 10 : 10, i.e. equal
- (D) 100 : 60 cannot be simplified

37. A car covers 60 km in the 1st hour, 60 km in the 2nd hour, and 60 km in the 3rd hour, all along the same straight highway. Its motion is:

- (A) non-uniform, because it travels a long way
- (B) uniform, because it covers equal distances in equal times
- (C) non-uniform, speeding up
- (D) circular, because it is on a highway

38. A train of length 150 m passes a stationary pole completely in 6 s. Its speed is:

- (A) 6 m/s
- (B) 900 m/s
- (C) 0.04 m/s
- (D) 25 m/s

39. A reading lamp's pendulum-style metronome arm swings to-and-fro about its pivot, repeating each beat at a fixed interval. This motion is:

- (A) both oscillatory and periodic
- (B) oscillatory but not periodic
- (C) periodic but not oscillatory
- (D) rotational

40. A bus driver wants a 120 km trip to average 60 km/h. The first 80 km took 2 hours. How long can the remaining 40 km take if she is to just meet the 60 km/h average?

- (A) 1 hour
- (B) 40 minutes
- (C) 0 hours — it is impossible without instant travel
- (D) 30 minutes

41. On a distance-time graph, object M's line is steeper than object N's line over the same time interval, both straight. Which statement is correct?

- (A) N covers more distance than M in the same time
- (B) both cover equal distance because the time is equal
- (C) M is at rest while N moves
- (D) M covers more distance than N in the same time

42. A car travels 100 km, then is stuck in traffic (at rest) for 1 hour, then travels another 100 km. The two moving stretches took 2 h and 2 h. The average speed for the entire trip, including the traffic, is:

- (A) 40 km/h
- (B) 50 km/h
- (C) 200 km/h
- (D) 60 km/h

43. On a distance-time graph, an object's line rises steeply, then less steeply, while always still rising. Over this time the object is:

- (A) moving backward toward the start
- (B) still moving forward, but slowing down
- (C) speeding up the whole time
- (D) at rest after the bend

44. A pendulum makes 40 complete oscillations in 50 s. How many complete oscillations will it make in 2 minutes?

- (A) 80
- (B) 96
- (C) 160
- (D) 48

45. Three runners: P runs at 6 m/s, Q covers 1.8 km in 5 minutes, and R runs at 25.2 km/h. Who is the fastest?

- (A) P, at 6 m/s
- (B) R, at 7 m/s
- (C) Q, at 6 m/s
- (D) all three are equal

46. A sand timer (hourglass) is trusted to measure a fixed interval because:

- (A) sand trickles through the narrow neck at a roughly steady rate
- (B) the sand grains repel each other periodically
- (C) the glass bulbs swing like a pendulum
- (D) sunlight heats the sand on a fixed schedule

47. A journey has three legs: 60 km at 60 km/h, then 60 km at 30 km/h, then 120 km at 40 km/h. The average speed for the whole journey is:

- (A) 43.3 km/h
- (B) 45 km/h
- (C) 60 km/h
- (D) 40 km/h

48. On a distance-time graph, a line that is perfectly horizontal corresponds to a speed of:

- (A) maximum speed
- (B) a very high constant speed
- (C) a speed that cannot be found
- (D) zero (the object is at rest)

49. A high-speed train runs at 180 km/h. How far does it travel in 30 seconds?

- (A) 5400 m
- (B) 6 m
- (C) 5000 m
- (D) 1500 m

50. A car covers a route in three equal-distance legs of 60 km each, at 20 km/h, 30 km/h and 60 km/h. The average speed for the whole 180 km is:

- (A) 36.7 km/h
- (B) 40 km/h
- (C) 30 km/h
- (D) 110 km/h

51. Which statement about distance-time graphs is FALSE?

- (A) A steeper straight line means a faster object
- (B) A straight slanted line means uniform speed
- (C) The slope of the line gives the speed
- (D) A horizontal line means the object moves at a steady high speed

52. A swimmer covers 1.2 km in 20 minutes at a steady pace. Her average speed in m/s is:

- (A) 60 m/s
- (B) 0.06 m/s
- (C) 3.6 m/s
- (D) 1 m/s

53. On a distance-time graph (distance measured from the start), an object's line rises to a peak and then falls back down to the time axis (distance 0). This means the object:

- (A) moved away and then returned to its starting point
- (B) kept speeding up the entire time
- (C) moved in a perfect circle
- (D) stopped permanently at the peak

54. A boy walks to school 2.4 km away at 6 km/h, then walks home along the same road at 4 km/h. His average speed for the round trip is:

- (A) 5 km/h
- (B) 4.8 km/h
- (C) 2.4 km/h
- (D) 10 km/h

55. A pendulum clock keeps perfect time. Its pendulum is then accidentally made shorter. From then on, compared with correct time, the clock will:

- (A) run slow, because a shorter pendulum swings less often
- (B) keep correct time, since period depends on the bob's mass
- (C) run fast (gain time), because a shorter pendulum has a shorter period
- (D) stop, because a short pendulum cannot swing

56. Two cars start from the same town. On a distance-time graph, car A's line is steeper than car B's, and both are straight from the origin. After any given time has passed:

- (A) car B is always farther because its line is gentler
- (B) they are always the same distance apart from each other
- (C) car A is behind car B until the lines cross
- (D) car A is always farther from the town than car B

57. A train 120 m long crosses a bridge 480 m long completely in 24 s. Its speed is:

- (A) 20 m/s
- (B) 5 m/s
- (C) 25 m/s
- (D) 30 m/s

58. Spot the INCORRECT speed conversion among these:

- (A) 36 km/h = 10 m/s
- (B) 54 km/h = 15 m/s
- (C) 72 km/h = 20 m/s
- (D) 126 km/h = 30 m/s

59. An odometer reads 12,460 km at the start of a drive and 12,700 km at the end; the drive lasted 4 hours. The car's average speed for the drive was:

- (A) 60 km/h
- (B) 3175 km/h
- (C) 240 km/h
- (D) 50 km/h

60. On a distance-time graph an object's straight line passes through (2 s, 10 m) and (7 s, 40 m). Its speed is:

- (A) 5 m/s
- (B) 8 m/s
- (C) 6 m/s
- (D) 30 m/s

Solutions — Science (Class 7), Chapter 13:

Motion and Time — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	B	31	A	41	D	51	D
2	B	12	B	22	A	32	A	42	A	52	D
3	B	13	D	23	A	33	B	43	B	53	A
4	D	14	B	24	C	34	C	44	B	54	B
5	D	15	B	25	A	35	B	45	B	55	C
6	B	16	A	26	C	36	A	46	A	56	D
7	C	17	D	27	B	37	B	47	D	57	C
8	C	18	C	28	C	38	D	48	D	58	D
9	A	19	A	29	A	39	A	49	D	59	A
10	D	20	D	30	C	40	C	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The tip moves along a circle, so the path is circular, not straight (rectilinear) or back-and-forth (oscillatory). Because it returns to the same spot after a fixed 60 s interval and repeats, the motion is periodic. It is not random and not oscillatory, so 'circular and periodic' is the only fit.

2. (B) Speed = distance / time = $240 / 16 = 15$ m/s. The option 16 m/s wrongly reports the time as the speed; 0.067 m/s divides time by distance ($16/240$) the wrong way; 3840 multiplies distance by time. So 15 m/s is correct.

3. (B) To go from km/h to m/s, multiply by $5/18$: $108 \times 5/18 = 540/18 = 30$ m/s. The value 388.8 multiplies by 3.6 (the wrong direction); 60 doubles the answer; 10.8 just shifts a decimal. So 30 m/s is correct.

4. (D) Total distance = $12 + 12 = 24$ km. Total time = $0.5 \text{ h} + 0.5 \text{ h (rest)} + 1 \text{ h} = 2 \text{ h}$. Average speed = $24 / 2 = 12$ km/h. Forgetting the rest gives $24/1.5 = 16$ km/h; averaging

the two leg-speeds (24 and 12) gives 18 km/h; ignoring all stops gives 24. The rest must be counted, so 12 km/h is correct.

5. (D) Take each half as 60 km. First half time = $60/30 = 2$ h; second half time = $60/60 = 1$ h. Total distance = 120 km, total time = 3 h, average = $120/3 = 40$ km/h. The plain mean $(30+60)/2 = 45$ km/h is the classic trap; 50 and 90 are unrelated. Equal distances mean more time is spent slow, pulling the average below 45, so 40 km/h is correct.

6. (B) The time period of a simple pendulum depends only on the length of the string, not on the mass of the bob and not on the amplitude (release angle). Same length means the same period regardless of iron vs wood, so the two are equal. Mass and amplitude do not change the period, ruling out the other three.

7. (C) A speedometer shows the instantaneous speed at that moment (the '80'), while an odometer accumulates and shows the total distance travelled (the number that climbed by 45 km). The pairing must be speedometer-then-odometer; swapping them is the trap, and a clock measures time, not either of these readings.

8. (C) To go from m/s to km/h, multiply by 3.6: $25 \times 3.6 = 90$ km/h. Dividing by 3.6 ($25/3.6 = 6.94$) goes the wrong way; 75 multiplies by 3; 45 uses 5/18 (the km/h-to-m/s factor). So 90 km/h is correct.

9. (A) Total distance = $400 + 100 = 500$ m. Total time = $50 + 100 = 150$ s. Average speed = $500 / 150 = 3.33$ m/s. Averaging the two leg-speeds (8 m/s and 1 m/s) gives 4.5 m/s, a trap; 5 and 1 are individual hints, not the whole-trip average. So about 3.3 m/s is correct.

10. (D) A horizontal line on a distance-time graph means the distance from the start stays at 40 m while time passes, so the object is stationary (speed = 0). A flat line never means faster; the value 40 is a position, not a speed; and the distance never decreases, so it is not moving backward. Hence it is at rest.

11. (C) Distance = $80 \times 1 + 40 \times 3 = 80 + 120 = 200$ km. Total time = $1 + 3 = 4$ h. Average = $200 / 4 = 50$ km/h. The plain mean $(80+40)/2 = 60$ km/h ignores that 3 hours were spent slow; 55 and 120 are unrelated. So 50 km/h is correct.

12. (B) Spinning about a fixed axis while staying in place is rotational motion. Rectilinear and translational motion mean moving along a straight path, which the bit does not; oscillatory means swinging to-and-fro, which it also does not. Hence rotational motion.

13. (D) Speed = slope = distance / time. P: $24/4 = 6$ m/s; Q: $16/4 = 4$ m/s. The steeper line P is faster, by $6 - 4 = 2$ m/s. Reaching $t = 4$ s together does not make them equal in speed (different distances), and the difference is 2 m/s, not 8. So P at 6 m/s vs Q at 4 m/s is correct.

14. (B) Any repeating (periodic) event marks equal intervals of time, which is exactly how clocks work. The clicks are evenly spaced only because the turning is periodic and steady; non-uniform or random motion would give uneven gaps, and an event that repeats at fixed intervals is by definition periodic. So 'periodic' is correct.

15. (B) First convert: $90 \text{ km/h} \times 5/18 = 25 \text{ m/s}$. Then distance = speed $\times$ time = $25 \times 8 = 200 \text{ m}$. Using 90 m/s by mistake gives 720 m ; 25 m forgets to multiply by the 8 s ; 180 multiplies 90 km/h 's number badly. So 200 m is correct.

16. (A) For a simple pendulum the time period depends only on the string length, not on how far back it is pulled (the amplitude). The bigger arc is covered at higher speed, so the period stays the same. Bigger swing does not lengthen or shorten the period, and extra energy does not change it either. Hence 'the same'.

17. (D) Speed = slope = change in distance / change in time = $(35 - 5) / (6 - 1) = 30 / 5 = 6 \text{ m/s}$. Reading only the first point (5 m at $1 \text{ s} = 5 \text{ m/s}$) ignores the slope; $35/6$ (about 5.8 , near 6) tempts a guess of 7 ; 30 is just the distance change. So 6 m/s is correct.

18. (C) Each hour covers a different distance: $36 + 54 + 72 = 162 \text{ km}$ in 3 h . Average = $162 / 3 = 54 \text{ km/h}$. Because the segments are equal in time, the average is simply the mean of the speeds here, 54 km/h . The value 162 is the total distance, not a speed; 48 and 60 are unrelated. So 54 km/h is correct.

19. (A) Time = distance / speed = $360 / 20 = 18 \text{ s}$. Convert speed: $20 \times 3.6 = 72 \text{ km/h}$. The pairing 18 s with 72 km/h is the only fully correct one; 56 km/h misconverts, 20 s misreads time, and 7200 s multiplies instead of dividing. So 18 s and 72 km/h .

20. (D) First convert distance: $9 \text{ cm} = 0.09 \text{ m}$. Speed = $0.09 / 3 = 0.03 \text{ m/s}$. Forgetting to convert cm to m gives 3 (cm/s) mislabelled; 0.3 misplaces a decimal; 27 multiplies instead of dividing. So 0.03 m/s is correct.

21. (B) On a distance-time graph the slope (steepness) equals speed, so the steeper line, X, has the greater speed. A steeper line means more distance per unit time, not less, so X is not slower and has not necessarily travelled less; both start at the origin, so neither started later. Hence X is faster.

22. (A) $5 \text{ minutes} = 300 \text{ s}$. Number of oscillations = total time / time period = $300 / 2 = 150$. Using 5 minutes directly without converting ($5/2 = 2.5$) is a trap; forgetting to divide by 2 gives 300 ; multiplying wrongly gives 75 . So 150 is correct.

23. (A) The lines meeting at $(4 \text{ s}, 20 \text{ m})$ means at $t = 4 \text{ s}$ both objects are 20 m from the start, i.e. at the same position. Crossing says nothing about their speeds (their slopes can differ), and neither stops nor reverses just because the lines meet. So they are at the same distance from the start.

24. (C) Time period = total time / number of oscillations. Before: $36 / 24 = 1.5$ s. After: $18 / 24 = 0.75$ s. So the period halved (1.5 to 0.75 s), which fits the shorter string giving a shorter period. It did not double or stay the same, and 24 is a count, not a period. So it halved to 0.75 s.

25. (A) km/h to m/s means multiply by $5/18$: $18 \times 5/18 = 5$ m/s. Multiplying by 3.6 ($18 \times 3.6 = 64.8$) goes the wrong way; 0.5 and 50 just shift the decimal. So 5 m/s is correct.

26. (C) Whole trip should take $300 / 75 = 4$ h. First half took $150 / 50 = 3$ h, leaving only 1 h for the last 150 km. Required speed = $150 / 1 = 150$ km/h. Simply averaging the deficit (100) or repeating 75 ignores the time already used up. So 150 km/h is needed.

27. (B) For the same distance, speed is inversely proportional to time, so the car with less time (A, 4 h) is faster. Speed ratio $A : B = (1/4) : (1/5) = 5 : 4$. Same distance does not mean same speed when times differ, and B (more time) is slower. So A is faster, 5 : 4.

28. (C) A straight slanted line means equal distances in equal times, i.e. uniform motion. Speed = $12 / 3 = 4$ m/s. A straight line is never non-uniform; 12 is the distance, not the speed; and a sloped line is not at rest. So uniform motion at 4 m/s.

29. (A) The distance from the start only ever increases (or pauses); it never comes back down, so the final reading, 70 m, is the total distance travelled. Adding all the segment values ($40 + 0 + 30$ again over the same path) double-counts; 110 and 40 misread the curve. So 70 m.

30. (C) Steepness on a distance-time graph means speed, so a line that keeps getting steeper means the speed keeps increasing, i.e. speeding up. A bending-up curve is not slowing down; a constant speed would be a straight line; and a rising line is not at rest. So it is speeding up.

31. (A) Sprinter: $150 / 15 = 10$ m/s. Horse: $600 \text{ m in } 60 \text{ s} = 600 / 60 = 10$ m/s. Both are 10 m/s, so they are equally fast. Comparing 150 directly with 600 (ignoring the different times) wrongly favours the horse; the 590 figure subtracts raw distances. So they are equal at 10 m/s.

32. (A) Average speed = total distance / total time, so you need the distance from the odometer and the time from a clock. A speedometer only shows the speed at one instant, which is not the trip average; a clock alone gives no distance. So the odometer plus a clock is correct.

33. (B) The distances covered each second fall: 6, then 4, then 2 m. Unequal distances in equal times means non-uniform motion, and since the distance per second is decreasing, the car is slowing down. Continuing forward does not make it uniform, the distances are not rising (not speeding up), and a moving toy is not at rest. So non-uniform, slowing down.

34. (C) Distance = speed x time = $300,000 \times 4 = 1,200,000$ km. Dividing ($300,000 / 4 = 75,000$) is the wrong operation; 300,000 ignores the 4 s; 1,200 drops three zeros. So 1,200,000 km.

35. (B) A longer pendulum has a longer time period, so it takes more time per swing and fits fewer swings into a minute. The shorter pendulum A has the smaller period, so it oscillates more often in the same minute. Length, not mass or energy, sets the count, and the longer one is slower. So A makes more.

36. (A) Speed A = $100 / 10 = 10$ m/s; Speed B = $60 / 10 = 6$ m/s. Ratio A : B = $10 : 6 = 5 : 3$. The reciprocal 3 : 5 reverses it; equal times do not mean equal speeds; and $100 : 60$ does simplify (to 5 : 3). So 5 : 3.

37. (B) Equal distances (60 km) in equal times (1 h each) is the definition of uniform motion; the speed is a steady 60 km/h. A long total distance does not make it non-uniform, the speed never changes (not speeding up), and a straight highway is rectilinear, not circular. So uniform.

38. (D) To pass a pole, the whole train length must go by, so distance = 150 m in 6 s. Speed = $150 / 6 = 25$ m/s. The value 6 reports the time as speed; 900 multiplies 150×6 ; 0.04 divides time by distance. So 25 m/s.

39. (A) Swinging to-and-fro is oscillatory, and because each beat repeats at a fixed time interval it is also periodic. Every oscillatory motion that repeats regularly is periodic, so 'oscillatory but not periodic' is wrong; it is not pure rotation (it does not spin all the way round). So both oscillatory and periodic.

40. (C) The whole trip should take $120 / 60 = 2$ h. But the first 80 km already used 2 h, leaving 0 h for the last 40 km. Covering 40 km in 0 time is impossible, so she cannot meet the average. Any positive time (1 h, 40 min, 30 min) would push the total over 2 h. So it is impossible.

41. (D) A steeper distance-time line means a higher speed, which means more distance in the same time. So M, the steeper line, covers more distance than N. Equal time intervals do not force equal distance when slopes differ, and a sloped line is not at rest. So M covers more.

42. (A) Total distance = $100 + 100 = 200$ km. Total time = $2 \text{ h} + 1 \text{ h (traffic)} + 2 \text{ h} = 5 \text{ h}$. Average = $200 / 5 = 40$ km/h. Leaving out the traffic hour gives $200/4 = 50$ km/h, the trap; 200 is the distance; 60 is unrelated. The idle time counts, so 40 km/h.

43. (B) The line keeps rising, so distance keeps increasing and the object moves forward. But the slope is getting smaller, and a smaller slope means a lower speed, so it is slowing down. A rising line is not moving backward, a flattening slope is not speeding up, and the line is not horizontal (not at rest). So forward but slowing down.

44. (B) Time period = $50 / 40 = 1.25$ s per oscillation. In 2 minutes = 120 s, number of oscillations = $120 / 1.25 = 96$. Using 100 s instead of 120 gives 80; doubling 80 gives 160; halving 96 gives 48. So 96.

45. (B) Put all in m/s. $P = 6$ m/s. $Q = 1800 \text{ m} / 300 \text{ s} = 6$ m/s. $R = 25.2 \times 5/18 = 126/18 = 7$ m/s. R at 7 m/s is the fastest; P and Q tie at 6 m/s but neither leads. So R is fastest.

46. (A) An hourglass works because the sand falls through its narrow neck at an almost constant rate, so a fixed amount of sand always takes the same time to run through. The grains do not pulse, the bulbs do not swing, and no sunlight is involved. So the steady trickle of sand is the reason.

47. (D) Times: $60/60 = 1$ h; $60/30 = 2$ h; $120/40 = 3$ h. Total distance = $60 + 60 + 120 = 240$ km; total time = $1 + 2 + 3 = 6$ h. Average = $240 / 6 = 40$ km/h. The plain mean of the leg speeds, $(60 + 30 + 40)/3 = 43.3$ km/h, is the trap; 45 and 60 are unrelated. So 40 km/h.

48. (D) A horizontal distance-time line means the distance does not change as time passes, so the object covers no ground: its speed is zero. A flat line never means maximum or high speed, and the speed is perfectly well-defined as 0. So zero.

49. (D) Convert: $180 \times 5/18 = 50$ m/s. Distance = $50 \times 30 = 1500$ m. Using 180 m/s gives 5400 m; dividing $180/30$ gives 6; 5000 is a slip. So 1500 m.

50. (C) Times: $60/20 = 3$ h; $60/30 = 2$ h; $60/60 = 1$ h. Total distance = 180 km; total time = $3 + 2 + 1 = 6$ h. Average = $180 / 6 = 30$ km/h. The plain mean $(20+30+60)/3 = 36.7$ km/h is the trap; equal distances spend more time at the slow speeds, pulling the average down to 30 km/h.

51. (D) A horizontal line means the distance is unchanging, so the object is at rest (speed zero), not moving fast — that statement is false. The other three are all true: steeper means faster, a straight slant means uniform speed, and the slope equals speed. So the false one is the 'horizontal = high speed' claim.

52. (D) Convert: $1.2 \text{ km} = 1200 \text{ m}$; $20 \text{ min} = 1200 \text{ s}$. Speed = $1200 / 1200 = 1$ m/s. Forgetting to convert minutes to seconds ($1200 \text{ m} / 20$) gives 60; misplacing the decimal gives 0.06; 3.6 is the km/h conversion of 1 m/s, not the answer asked. So 1 m/s.

53. (A) Distance falling back to zero means the object's distance from the start became 0 again, i.e. it came back to where it began. A line that rises then falls is not continuous speeding up, says nothing about a circular path, and the object kept moving after the peak (it did not stop). So it returned to its starting point.

54. (B) Going: $2.4 / 6 = 0.4$ h. Returning: $2.4 / 4 = 0.6$ h. Total distance = 4.8 km, total time = 1.0 h, average = $4.8 / 1.0 = 4.8$ km/h. The plain mean $(6+4)/2 = 5$ km/h is the trap;

2.4 is one leg's distance; 10 adds the speeds. So 4.8 km/h.

55. (C) A shorter pendulum has a shorter time period, so it ticks more often than it should and the clock gains time (runs fast). A shorter pendulum actually swings more often, not less; the period depends on length, not the bob's mass; and a short pendulum still swings fine. So it runs fast.

56. (D) Both lines start together at the origin; the steeper line (A) gains distance faster, so at every later moment A is ahead of B and the gap keeps widening. A gentler line is slower (not farther), the gap is not constant, and from the origin A is never behind B. So A is always farther.

57. (C) To cross completely the train travels its own length plus the bridge: $120 + 480 = 600$ m in 24 s. Speed = $600 / 24 = 25$ m/s. Using only the bridge ($480/24 = 20$) forgets the train length; using only the train ($120/24 = 5$) forgets the bridge; 30 is a slip. So 25 m/s.

58. (D) Multiply km/h by $5/18$: $126 \times 5/18 = 630/18 = 35$ m/s, not 30, so that conversion is wrong. Checking the rest: $36 \times 5/18 = 10$, $54 \times 5/18 = 15$, $72 \times 5/18 = 20$ — all correct. So the incorrect one is $126 \text{ km/h} = 30 \text{ m/s}$.

59. (A) Distance = $12,700 - 12,460 = 240$ km. Average speed = total distance / total time = $240 / 4 = 60$ km/h. Dividing the final reading by 4 ($12,700/4 \approx 3175$) ignores the start; 240 forgets to divide by time; 50 is a slip. So 60 km/h.

60. (C) Speed = slope = change in distance / change in time = $(40 - 10) / (7 - 2) = 30 / 5 = 6$ m/s. Reading the first point alone (10 m at 2 s = 5 m/s) ignores the slope; $40/5 = 8$ uses the wrong distance; 30 is just the distance change. So 6 m/s.